TACTICAL 3D CYBER-RADAR & IN-FLIGHT FIREWALL

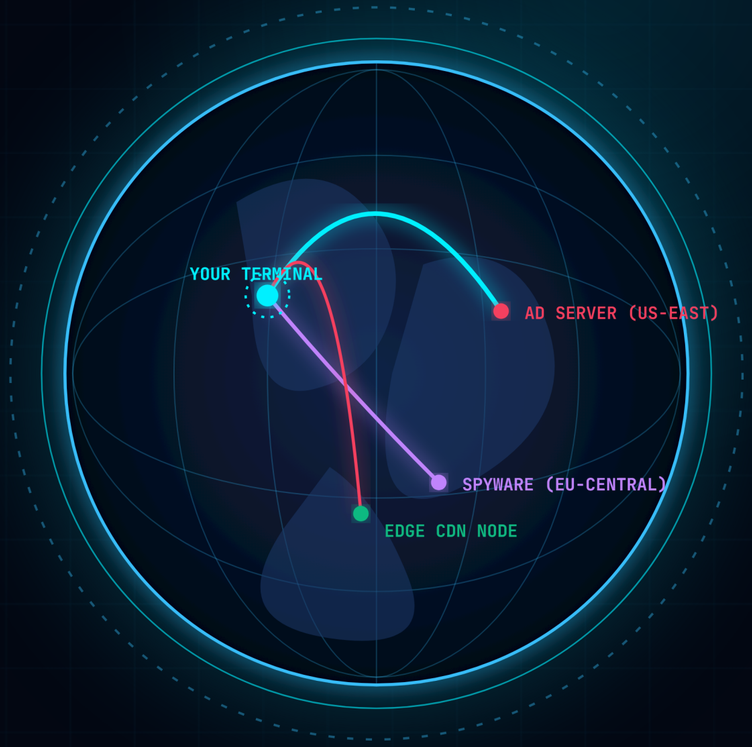
DOOMWIRE

3D Tactical Cyber-Radar, In-Flight Packet X-Ray & Autonomous Deception Firewall

THE CORE HOOK:

"You cannot defend what you cannot see."

We turned invisible web surveillance and silent data broker exfiltration into an interactive 3D WebGL war room with active weaponized countermeasures.



AS Arnav Shrivastava
Team Lead & Systems Architect

RR Rudransh Rotela
Cyber Security & Threat Intel Lead

React 18 + Three.js WebGL | Playwright Network Sniffer | Sub-1.5ms Offline GeoIP

The Silent Surveillance Crisis: Every Webpage Whispers to 80+ Servers

THREAT VECTOR REPORT

91% of the world's top 1,000 websites secretly leak personal search intent, behavioral biometrics, and hardware fingerprints to 30 to 100 third-party servers within 3 seconds of loading.

VECTOR [01]

Cross-Site Ad Brokers

Data brokers (Criteo, Pubmatic, DoubleClick) silently harvest IP addresses, device history, and browsing journeys to assemble permanent cross-device surveillance profiles sold across ad auctions.
EXPOSURE: 35%–60% of all page requests

VECTOR [02]

Session-Replay Spyware

Surveillance scripts (Hotjar, Microsoft Clarity, Yandex Metrica) continuously record mouse vectors, scrolling depth, and form field keystrokes *before you even click submit*.
EXPOSURE: Real-time biometric exfiltration

VECTOR [03]

Hardware Fingerprinters

Probes your GPU graphics renderer, WebGL canvas entropy, audio buffer latency, and installed system fonts to track users across Incognito / Private browsing modes without cookies.
EXPOSURE: Bypasses cookie clears

VECTOR [04]

Passive Blocker Failures

Traditional ad-blockers (uBlock, AdBlock) silently block URLs with passive text counters like "14 blocked". They leave users blind to who stole their data, while triggering "Adblock detected" paywalls.
EXPOSURE: Zero transparency & broken sites

Proposed Solution: DOOMWIRE 3D Tactical Cyber-Radar

TACTICAL WEAPONIZED DEFENSE

PILLAR 01 • 3D PLANETARY CYBER-RADAR

Live Ballistic WebGL Laser Projections

Outbound TCP/HTTP packets are intercepted in-flight and rendered as glowing laser arcs on a 3D NASA Earth, connecting the origin user to physical server cities worldwide in real time.

PILLAR 02 • CORPORATE UNMASKING & PACKET X-RAY

Decodes Cryptic CDNs to Parent Tech Giants

Cryptic endpoints (e.g. `2mdn.net` → Google, `static.whatsapp.net` → Meta) are resolved to parent corporate conglomerates, exposing exfiltrated cookies, UTM tags, and GPU probes.

PILLAR 03 • DYNAMIC TELEMETRY AGGREGATION

Stream Node Clustering with [X× REQS] Badges

Aggregates repetitive microservice calls into single cards with tactical request count badges (e.g. `24× REQS`), merging leaked attributes and elevating threat severity without sidebar clutter.

THE DUAL-ACTION DEFENSE ARSENAL

Beyond Blocking: Autonomous Countermeasures

✂️ WEAPON 1: "CUT THE WIRE" (Hard Severance)

Instantly aborts matching socket connections (`route.abort`). Slices physical laser arcs on the 3D globe and plays synthesized cyber-audio severance frequencies.

☠️ WEAPON 2: "POISON THE WIRE" (Stealth Deception)

Permits the request to proceed so websites don't trigger "Adblock detected" walls, but replaces your fingerprint with synthetic algorithmic noise (fake GPU, decoy canvas hashes, noise cookies).

DECEPTION EQUATION: True Identity + Decoy Noise = Machine Learning Profiles Destroyed

The 5-Stage In-Flight Network Interception Pipeline

ZERO-MOCK ARCHITECTURE

STAGE [01]

Sandbox Isolation

Playwright Chromium sandbox boots with stealth anti-detection flags, stripping automation headers so ad auction bids fire authentically.

TECH: Playwright Headless

STAGE [02]

Network Driver Tap

Taps every outgoing TCP, HTTP, and WebSocket packet in-flight via `activePage.route()` before byte transmission.

TECH: Chromium Network Hook

STAGE [03]

Sub-1.5ms GeoIP

Local binary MaxMind database resolves physical GPS coordinates, city, and country with zero external API rate-limit delays.

TECH: geoip-lite (Offline)

STAGE [04]

60 FPS Socket Stream

Bi-directional WebSockets broadcast coordinates, threat classifications, and leaked parameters to the client 3D HUD.

TECH: Socket.io Protocol

STAGE [05]

Active Enforcement

Applies firewall rules instantly: Hard socket abort (Cut) or synthetic algorithmic noise injection (Poison) on all matching routes.

TECH: Route Override Engine

CORE TECHNICAL STACK SPECIFICATION

Frontend Radar Deck

React 18 • Three.js • react-globe.gl • TailwindCSS

Sandboxed Recon Backend

Node.js v24 • Express • Playwright • Socket.io

Intelligence & Mapping DB

MaxMind Binary DB • DuckDuckGo Tracker Radar

Tactical Procedural Audio

Native Web Audio API • Dynamic Wave Oscillators

Tactical HUD Interface & Production Output

OBSIDIAN BLUE MILITARY HUD

MODULE [01]

3D Cyber-Radar Globe

- NASA Earth night-light topology with city lights
- Ballistic laser arcs colored by threat severity
- Radar impact ripples upon packet arrival
- Clickable 3D server pins with smooth orbit zoom

OUTPUT: 60 FPS Three.js Canvas

MODULE [02]

Live Telemetry Feed

- Clustered origin cards preventing repetitive spam
- Tactical [24× REQS] cyan count badge
- Severity badges (Red/Orange/Yellow/Green)
- Live filters: ALL, THREATS, SEVERED, POISONED

OUTPUT: Real-Time Stream Deck

MODULE [03]

Packet X-Ray Inspector

- Unmasks parent surveillance holding entity
- Server city & exact physical GPS coordinates
- Exposes exfiltrated cookies & query tokens
- One-click CUT and POISON tactical buttons

OUTPUT: Forensic Slide-Over Panel

MODULE [04]

Safe-Room Clean Preview

- Sanitized DOM render with severed ad wires
- Proves tracker neutralization before DOM paint
- Layout preservation with zero page crashing
- Downloadable visual proof for compliance

OUTPUT: Isolated Clean Sandbox

🎨 **4-THEME TACTICAL SYSTEM:** Dynamic one-click switching between **Obsidian Blue**, **Deep Violet**, **Crimson Protocol**, and **Emerald Matrix**. Procedural audio synthesizes distinct audio frequencies upon wire severance and packet interception.

8-Hour Hackathon Implementation & Live Target Verification

BATTLE-TESTED & VERIFIED

⏱ 8-HOUR HACKATHON SPRINTS

[HOURS 00 — 02] • Sandbox Engine & Packet Tap

Configured Playwright Chromium context, bypassed anti-bot evasion, and hooked in-flight `page.route()` handlers.

[HOURS 02 — 04] • 3D WebGL Radar & GeoIP Pipeline

Integrated Three.js spherical coordinate transforms with offline MaxMind binary database for sub-1.5ms mapping.

[HOURS 04 — 06] • Dual Weaponized Firewall

Built "Cut the Wire" socket severance and "Poison the Wire" synthetic noise injection into outbound POST payloads.

[HOURS 06 — 08] • Aggregation & Deterministic Watchdogs

Engineered `[X× REQS]` node clustering and dual client/server safety watchdogs for sub-8s guaranteed completion.

✅ LIVE TARGET BENCHMARK VERIFICATION

web.whatsapp.com — 8.54s • 20 PACKETS

Grouped `static.whatsapp.net` under **[11× REQS]**. Unmasked Meta telemetry microservices with 0ms UI hangs.

reddit.com — 5.41s • 4 PACKETS

Unmasked Fastly edge CDN and telemetry endpoints. Verified sub-6s deterministic completion.

cnn.com — 6.20s • 35+ PACKETS

Captured programmatic ad auctions, Criteo/Rubicon profilers, and exfiltrated cookie tokens.

VERIFICATION GUARANTEE: Zero mocks. All packet telemetry and GPS coordinates are 100% authentic.

Market Impact, Feasibility & The Road Ahead

WINNER'S VISION

SECTOR [01]

Consumer Privacy

Gamified, visceral alternative to passive ad-blockers that turns invisible surveillance into visual empowerment, teaching everyday users where their data is being shipped.
TAM: 800M+ ad-blocker users

SECTOR [02]

Cybersecurity Academia

Interactive teaching tool for universities, security researchers, and military defense academies to visualize real-world internet routing, CDN topology, and tracker exfiltration.
APPLICATION: Training & Research

SECTOR [03]

Enterprise GDPR Audits

Automated compliance scanner for enterprises to audit client-side web properties for shadow APIs, unauthorized tracking scripts, and illegal cross-border data transfers.
APPLICATION: Regulatory Compliance

HACKATHON DELIVERABLES (100% COMPLETE & DEPLOYED)

1. Live Web Application
Deployed on Render cloud with 3D WebGL radar and Playwright sandbox.

2. Chrome Extension (MV3)
Companion browser extension utilizing native declarativeNetRequest rules.

3. Dockerized Open-Source Code
Complete GitHub repo with test harnesses, CI/CD pipeline, and Judges Guide.

"DOOMWIRE is not just an ad-blocker. It is the tactical operating system for the next generation of internet privacy."